

SF1811 Optimization

Lecture 6: Lagrangian relaxation and sensitivity

Chapters 6 and 22

Anders Forsgren

Lagrangian relaxation and sensitivity

Main question:

How does Lagrangian relaxation generate the dual problem and explain the meaning of the multipliers?

Today:

- Relaxation and lower bounds.
- Equality and inequality constraints.
- The Lagrangian dual problem.
- The dual of a linear program.
- Global optimality conditions.
- Sensitivity and the interpretation of multipliers.

Relaxation

Consider

$$(P) \quad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in S, \end{array} \qquad (P_R) \quad \begin{array}{ll} \text{minimize} & f_R(x) \\ \text{subject to} & x \in S_R. \end{array}$$

Definition: Relaxation

The problem (P_R) is a relaxation of (P) if

$$S \subseteq S_R$$

and

$$f_R(x) \leq f(x), \qquad x \in S.$$

Proposition

The optimal values satisfy $\text{optval}(P_R) \leq \text{optval}(P)$.

When is a relaxation exact?

Assume that (P_R) is a relaxation of (P).

Proposition

Suppose that $\hat{x}$ is optimal to (P_R) and that

$$\hat{x} \in S, \qquad f_R(\hat{x}) = f(\hat{x}).$$

Then $\hat{x}$ is optimal to (P).

Proof

For every $x \in S$,

$$f(\hat{x}) = f_R(\hat{x}) \leq f_R(x) \leq f(x).$$

If the relaxed lower bound is attained by a feasible point without a gap, the original problem has been solved.

Equality and inequality constraints

Consider the problem

$$(P) \qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0, \quad i \in \mathcal{E}, \\ & g_i(x) \leq 0, \quad i \in \mathcal{I}, \\ & x \in X. \end{array}$$

Use multipliers

$$u_i \in \mathbb{R}, \qquad i \in \mathcal{E},$$

for the equality constraints, and

$$y_i \geq 0, \qquad i \in \mathcal{I},$$

for the inequality constraints.

Equality multipliers are free. Inequality multipliers have a sign restriction.

The Lagrangian

The Lagrangian is

$$L(x, u, y) = f(x) + u^{\mathsf{T}} h(x) + y^{\mathsf{T}} g(x),$$

where

$$u \in \mathbb{R}^{|\mathcal{E}|}, \qquad y \geq 0.$$

If x is feasible to (P), then

$$h(x) = 0, \qquad g(x) \leq 0.$$

Therefore,

$$u^{\mathsf{T}} h(x) = 0$$

for every u, while

$$y^{\mathsf{T}} g(x) \leq 0$$

when $y \geq 0$.

Hence, for every primal feasible x,

$$L(x, u, y) \leq f(x).$$

The Lagrangian relaxation problem

Fix multipliers $u \in \mathbb{R}^{|\mathcal{E}|}$ and $y \geq 0$.

The Lagrangian relaxation problem is

$$(P_{u,y}) \qquad \begin{array}{ll} \text{minimize} & L(x,u,y) \\ \text{subject to} & x \in X. \end{array}$$

The constraints

$$h(x) = 0, \qquad g(x) \leq 0$$

have been moved into the objective function.

The multipliers are fixed when solving the relaxation. The variable is x.

The dual function

Define the dual function

$$\varphi(u, y) = \min_{x \in X} L(x, u, y).$$

For every $u \in \mathbb{R}^{|\mathcal{E}|}$ and $y \geq 0$,

$$\varphi(u, y) \leq \text{optval}(P).$$

Thus, every admissible pair (u, y) gives a lower bound on the primal optimal value.

How should the multipliers be chosen to obtain the strongest lower bound?

The Lagrangian dual problem

The Lagrangian dual problem chooses the strongest lower bound:

$$(D) \qquad \begin{array}{ll} \text{maximize} & \varphi(u, y) \\ \text{subject to} & u \in \mathbb{R}^{|\mathcal{E}|}, \\ & y \geq 0. \end{array}$$

Proposition (Weak duality)

The optimal values satisfy $\text{optval}(D) \leq \text{optval}(P)$.

The difference

$$\text{optval}(P) - \text{optval}(D)$$

is called the duality gap.

For linear programming, the duality gap is zero when the primal problem has a finite optimal value.

A linear program as a constrained problem

Consider the standard-form linear program

$$(P) \qquad \begin{array}{ll} \text{minimize} & c^{\mathsf{T}} x \\ \text{subject to} & Ax = b, \\ & x \geq 0. \end{array}$$

To match the general form, write

$$h(x) = b - Ax = 0$$

and

$$g(x) = -x \leq 0, \qquad X = \mathbb{R}^n.$$

Use a free multiplier $y \in \mathbb{R}^m$ for $h(x) = 0$ and a multiplier $r \geq 0$ for $g(x) = -x \leq 0$.

The vector r will become the reduced-cost vector from the simplex method.

Lagrangian relaxation of the linear program

The Lagrangian is

$$\begin{aligned} L(x, y, r) &= c^{\mathsf{T}} x + y^{\mathsf{T}}(b - Ax) + r^{\mathsf{T}}(-x) \\ &= b^{\mathsf{T}} y + \left(c - A^{\mathsf{T}} y - r\right)^{\mathsf{T}} x, \end{aligned}$$

where

$$y \in \mathbb{R}^m, \qquad r \geq 0, \qquad x \in \mathbb{R}^n.$$

The dual function is

$$\varphi(y, r) = \min_{x \in \mathbb{R}^n} \left\{ b^{\mathsf{T}} y + \left(c - A^{\mathsf{T}} y - r\right)^{\mathsf{T}} x \right\}.$$

The dual function for the linear program

Since x is free in the Lagrangian relaxation,

$$\varphi(y, r) = \begin{cases} b^{\mathsf{T}} y, & A^{\mathsf{T}} y + r = c, \\ -\infty, & \text{otherwise.} \end{cases}$$

The Lagrangian dual problem is therefore

$$\begin{array}{ll} \text{maximize} & b^{\mathsf{T}} y \\ \text{subject to} & A^{\mathsf{T}} y + r = c, \\ & r \geq 0. \end{array}$$

which is precisely the dual problem from Lecture 5.

Why introduce r explicitly?

Eliminating r from

$$A^{\mathsf{T}} y + r = c, \qquad r \geq 0,$$

gives the equivalent condition

$$A^{\mathsf{T}} y \leq c.$$

However, keeping r explicit shows the connection to the simplex method:

$$r = c - A^{\mathsf{T}} y.$$

For a simplex basis,

$$r_B = 0, \qquad r_N = c_N - A_N^{\mathsf{T}} y.$$

It also makes the complementarity condition transparent:

$$x^{\mathsf{T}} r = 0.$$

The Lagrange multipliers for $-x \leq 0$ are the reduced costs.

Tables and chairs: resource prices

Recall the production problem

$$\begin{array}{ll} \text{maximize} & 400T + 300C \\ \text{subject to} & T + C \leq 200, \\ & 2T + C \leq 300, \\ & T, C \geq 0. \end{array}$$

Let

$$y_1 = \text{price of one big part}, \qquad y_2 = \text{price of one small part}.$$

A table uses one big and two small parts, while a chair uses one of each. For the resource offer to dominate production revenue, require

$$y_1 + 2y_2 \geq 400, \qquad y_1 + y_2 \geq 300.$$

The total resource value is

$$200y_1 + 300y_2.$$

Sensitivity to the right-hand side

Consider the standard-form linear program with a changed right-hand side:

$$A\widetilde{x} = b + \delta e_i, \qquad \widetilde{x} \geq 0.$$

Assume that the same basis remains primal and dual feasible. Then

$$\widetilde{x}_B = A_B^{-1}(b + \delta e_i), \qquad \widetilde{x}_N = 0.$$

The dual solution y and reduced costs r are unchanged, since they depend on A, c, and the basis, but not on b.

The new optimal value is

$$\begin{aligned}\widetilde{z} &= (b + \delta e_i)^{\mathsf{T}} y \\ &= b^{\mathsf{T}} y + \delta y_i.\end{aligned}$$

As long as the basis remains optimal, y_i is the change in optimal value per unit change in b_i.

General-form LP and the formula sheet

The formula sheet gives primal-dual optimality conditions for a linear program with

- inequality and equality constraints;
- nonnegative and free variables.

The sign and equality rules follow from Lagrangian relaxation:

$$\begin{aligned}
\text{primal equality} &\iff \text{free dual variable},\\
\text{primal inequality} &\iff \text{sign-restricted dual variable},\\
\text{free primal variable} &\iff \text{dual equality},\\
\text{sign-restricted primal variable} &\iff \text{dual inequality}.
\end{aligned}$$

Different LP forms contain the same primal-dual information.

Main ideas

After this lecture, you should be able to:

- explain why a relaxation gives a lower bound;
- distinguish free equality multipliers from sign-restricted inequality multipliers;
- formulate the Lagrangian, dual function, and Lagrangian dual problem;
- derive the dual of a standard-form linear program;
- identify the reduced costs as multipliers for $-x \leq 0$;
- explain why strong duality makes the best Lagrangian lower bound exact for linear programming;
- interpret dual variables through sensitivity and shadow prices.

Dual variables describe both optimality certificates and the marginal value of constraints.